

EDUCATION

Princeton University

Princeton, NJ

Advisors: Prof. Aleksandra Korolova and Prof. Jonathan Mayer

Aug. 2026 – Present

Ph.D. in Computer Science

University of Pennsylvania

Philadelphia, PA

Aug. 2022 – May 2026

B.S.E. in Artificial Intelligence GPA: 3.87/4.0

PUBLICATIONS

C2. Maya Pal Gambhir, Bailey Flanigan, and Aaron Roth. “Near Optimal Dropout-Robust Sortition.” *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026. **Spotlight**.

C1. Maya Pal Gambhir*, Maxon Rubin-Toles*, Keshav Ramji, Aaron Roth, and Surbhi Goel. “Conformal Language Model Reasoning with Coherent Factuality.” *International Conference on Learning Representations (ICLR)*, 2025. (*equal contribution)

RESEARCH EXPERIENCE

Princeton University, Center for Information Technology Policy

Princeton, NJ

Advisors: Prof. Aleksandra Korolova and Prof. Jonathan Mayer

Aug. 2026 – Present

Graduate Researcher

- Investigating the measurability of advertising effectiveness for small businesses, with a focus on causal inference approaches to estimating return on ad spend and the structural barriers that limit small business access to meaningful measurement tools.

University of Pennsylvania, Department of Computer Science

Philadelphia, PA

Advisors: Prof. Surbhi Goel and Prof. Bailey Flanigan (MIT)

Sep. 2025 – May 2026

Undergraduate Thesis Researcher

- Identified a compact set of pool- and quota-level statistics that predict optimal Maximin and Minimax marginal selection probabilities to within 1–2% mean absolute error across a corpus of 65 real-world and 6,000+ synthetic sortition instances.
- Proposed a privacy-preserving synthetic instance generation pipeline (exact quota replication + IPF second-order marginal matching + dependent rounding) to enable open benchmarking of citizens’ assembly selection algorithms without releasing sensitive demographic microdata.

University of Pennsylvania, Department of Computer Science

Philadelphia, PA

Advisors: Prof. Aaron Roth, Prof. Surbhi Goel, and Prof. Bailey Flanigan (MIT)

Jan. 2023 – May 2026

Undergraduate Research Assistant

- Developed conformal prediction methods for LLM reasoning chains that provide coherent factuality guarantees across multi-step outputs, accounting for inter-claim dependencies; achieved 90% factuality while retaining 80%+ of original model outputs. Published at ICLR 2025.
- Designed and analyzed a minimax optimization algorithm for dropout-robust sortition panel selection; proved low-regret convergence to the optimal value with high probability on sampled panels and validated on real-world data. Published at AISTATS 2026 (Spotlight).

University of Pennsylvania, Computational Social Science Lab

Philadelphia, PA

Supervisor: Baird Howland (advised by Prof. Duncan Watts)

Jan. 2023 – May 2023

Undergraduate Research Assistant

- Evaluated the ability of LLMs to detect and classify emotions elicited by news sources across a large text corpus; iteratively refined ChatGPT API prompts for quote extraction and analyzed error rates across models.

TEACHING EXPERIENCE

Teaching Assistant, University of Pennsylvania

CIS 5230 (Graduate-level): Ethical Algorithm Design

Spring 2026

CIS 3333: Mathematics of Machine Learning

Fall 2025

CIS 1600: Mathematical Foundations of Computer Science

Summer 2025

AWARDS AND HONORS

AI Academic Excellence Award, Raj and Neera Singh BSE in AI Program

2026

- Awarded to the top graduating student in Penn Engineering's undergraduate AI degree program.

Wolf-Hallac Award, Penn Engineering

2026

- Recognizes the outstanding graduating female senior from Penn Engineering (top 10% GPA, leadership, and community impact).

Computing Research Association Undergraduate Research Award, Honorable Mention

2025

- Recognizes North American undergraduates with outstanding computing research potential.

Rachleff Scholars Program, University of Pennsylvania

2022 – 2026

- Selected for a small cohort of Penn Engineering undergraduates; 10-week paid summer research fellowship with faculty mentorship.

SERVICE

Reviewer

International Conference on Learning Representations (ICLR)

2026

International Conference on Artificial Intelligence and Statistics (AISTATS)

2026